Assignment 2 — Management Dashboard

Course: Business Analytics and AI as a Tool for Effective Management
Program: MIM
Professor: Irina Nikivincze
Due Date: March 22, 2026

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1. Business Requirements & Data Preparation

Business Problem

The objective of this dashboard is to monitor Deloitte consulting project performance by evaluating the impact of Artificial Intelligence (AI) on efficiency, profitability, and client satisfaction. The dashboard helps determine whether AI adoption improves project outcomes and optimizes resource utilization.

Business Objectives

1. Increase project margins by at least 5%
2. Reduce average project duration by 15%
3. Improve client satisfaction scores to at least 4.5 (out of 5)
4. Reduce manual document review hours by at least 20%
5. Increase consultant strategic time allocation to at least 25%

Target Audience

The primary users of this dashboard are project managers, department heads, and analytics teams responsible for monitoring performance and making data-driven decisions.

Data Preparation

The dataset consists of 120 records representing consulting projects. Key variables include Project Type, Region, Industry, Project Duration, Project Margin, Client Satisfaction, Consultant Hours, Hours Saved, and Strategic Time.

Data cleaning involved removing duplicates, correcting inconsistencies, and ensuring proper data types. Calculated metrics were created, including Project Margin (%), Average Duration, Client Satisfaction, Hours Saved, and Strategic Time (%).

2. Dashboard Screenshot

3. Dashboard Link

https://disk.360.yandex.com/d/8Bf84-kUMnRNpw

4. Dashboard & Alarm Description

Key Performance Indicators (KPIs)

Project Margin (%): Measures project profitability.
🟢 Green indicates strong performance (≥ 8%)
🟡 Yellow indicates moderate performance (5% – 8%)
🔴 Red indicates low performance (< 5%)

Project Duration (Days): Tracks project delivery time.
🟢 Green indicates efficient delivery (≤ 70 days)
🟡 Yellow indicates moderate delay (70 – 90days)
🔴 Red indicates significant delay (> 90 days)

Client Satisfaction (CSAT): Reflects client feedback and service quality.
🟢 Green indicates high satisfaction (≥ 4.5)
🟡 Yellow indicates moderate satisfaction (3.5 – 4.5)
🔴 Red indicates low satisfaction (< 3.5)

Hours Saved: Represents efficiency gained from AI.
🟢 Green indicates high efficiency (≥ 20 hours)
🟡 Yellow indicates moderate efficiency (10 – 20 hours)
🔴 Red indicates low efficiency (< 10 hours)

Strategic Time (%): Shows time spent on high value activities.
🟢 Green indicates optimal allocation (≥ 60%)
🟡 Yellow indicates moderate allocation (40% – 60%)
🔴 Red indicates low allocation (< 40%)

Charts and Visualizations

- **Line Chart:** Displays project duration trends over time, comparing AI and non-AI projects.
- **Bar Chart:** Shows project margin comparison between AI and non-AI projects.
- **Heat Map (Matrix):** Illustrates client satisfaction across regions and industries using color gradients.
- **Donut Chart:** Represents the distribution of consultant time between manual and strategic activities.
- **Combo Chart:** Demonstrates the relationship between hours saved and project margin.

Filters and Interactivity

The dashboard includes interactive filters for:

- Project Type (AI / Non-AI)
- Region
- Industry
- Year

These filters allow users to explore data dynamically and focus on specific segments.

Alerts and Actions

The dashboard uses color coded indicators:

- Green indicates good performance
- Yellow indicates moderate performance
- Red indicates areas requiring attention

These alerts help users quickly identify issues and take actions such as reallocating resources, optimizing processes, or improving project strategies.

5. Peer Dashboard Review

Reviewer: Deloitte Group
Dashboard Reviewed: Monthly Sales Performance Dashboard
Dashboard Creator: Hafsa Idsaid (Group)

Summary & First Impressions

The dashboard presents an overview of monthly sales performance across different product categories and regions. It is designed for business users such as sales managers to monitor key metrics including total sales, profit, discount, and quantity. The layout is structured and provides a quick snapshot of overall performance.

Strengths

- The dashboard has a clear and organized structure, with KPI cards positioned at the top for quick visibility.
- Important metrics such as Total Sales, Total Profit, Average Discount, and Quantity are well presented.
- The use of multiple visualizations (bar chart, donut chart, line chart, and table) improves data understanding.
- Filters for year and month enhance interactivity and allow users to explore the data dynamically.
- Consistent use of color (blue theme) makes the dashboard visually cohesive.

Constructive Critique & Recommendations

- The dashboard lacks clear insights or explanations, making it harder to interpret key findings.
- Some visuals appear crowded, reducing readability and clarity.
- The donut chart provides distribution but lacks actionable insight.
- There is no use of color alerts (red, yellow, green) to highlight performance issues.
- Chart titles could be improved to be more insight-driven and descriptive.

Conclusion

Overall, the dashboard effectively presents sales data and demonstrates good use of visualization techniques. However, it would benefit from improved storytelling, clearer insights, and better use of visual alerts to enhance decision-making.